Uropharmacology

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Block: Endocrine/Urology/Reproductive
Thread: Pharmacology

Conflict of Interest: Dr. Trabulsi has financial interests/consulting with CG Oncology and Protara Therapeutics

Learning Objectives

By exam time, you should be able to do the following:

1. Describe the pharmacologic management of benign prostatic hyperplasia (BPH), including indications, medications, mechanisms of action, and side effects/drug interactions
2. Describe the pharmacologic management of erectile dysfunction (ED), including indications, medications, mechanisms of action, and side effects/drug interactions
3. Describe the pharmacologic management of overactive bladder (OAB) and lower urinary tract symptoms (LUTS), including indications, medications, mechanisms of action, and side effects/drug interactions
4. Describe the pharmacologic management of hypogonadism, including indications, medications, mechanisms of action, and side effects/drug interactions
5. Describe hormonal-based therapies for prostate cancer, including indications, medications, mechanisms of action, and side effects/drug interactions

Key Words and Drugs

Abiraterone
Acetylcholine
Alfuzosin
Alpha blocker
Alprostadil
Androgen deprivation
Antimuscarinic
Apalutamide
Bladder neck
BPH
cGMP
Corporal smooth muscle
Darolutamide
Darifenacin
Degarelix
Doxazosin
Dutasteride
ED
Enzalutamide
Fesoterodine
Finasteride
GnRH
Goserelin
Hyoscyamine
Hypogonadism
Intracavernosal
Leuprolide
LH

LUTS
MUSE
Mirabegron
Oxybutynin
OAB
Papaverine
PDE5i
Phentolamine
Phosphodiesterase 5 inhibitor
Prostatic Urethra
Prazosin
Relugolix
Sildenafil
Silodosin
Solifenacin
Tadalafil
Tamsulosin
Terazosin
Testosterone
Tolterodine
Vardenafil
Vibegron
5 Alpha reductase inhibitor
5ARI

Study Questions

1. A 75-year-old patient complains of worsening urinary symptoms, including nocturia, urgency, frequency, and rare urge urinary incontinence. Urinalysis is normal, and digital rectal exam reveals an enlarged prostate, which is soft without any induration or nodularity. The patient also has a cataract, and is scheduled for cataract surgery next month. The patient wishes to try a medication to help with voiding. Which of the following medications would be contra-indicated for this patient?
    a. Tadalafil
    b. Tolterodine
    c. Tamsulosin
    d. Oxybutynin
    e. Mirabegron

2. An 84-year-old nursing home resident is noted to have urinary incontinence. Urinalysis is normal, and the patient states "I can't get to the bathroom quick enough". A urodynamic evaluation reveals uninhibited bladder contractions consistent with overactive bladder. The nursing home staff put the patient in diapers, but the patient is very unhappy wearing them. The patient has had some memory loss over the past several years but is alert to person place and time. The most appropriate medication to try to alleviate the patient's symptoms is:
    a. Oxybutynin immediate release
    b. Trospium extended release
    c. Darifenacin
    d. Tolterodine
    e. Dutasteride

3. A 45-year-old patient complains of gradual onset of erectile dysfunction which has worsened since starting medication for hypertension. His internist has ruled out other causes, and referred him to the urologist for evaluation and treatment. He is single, and wishes to maintain spontaneity as much as possible. Which medication will be most appropriate for him?
    a. Sildenafil 50mg
    b. Tadalafil 20mg
    c. Vardenafil 10mg
    d. Papavarine 15mg
    e. Tadalafil 5mg

4. You are asked to see a 62-year-old patient in the ER for urinary retention. He is a poor historian, but does state he was taking 2 different pills for his "prostrate", the names of which he cannot remember. On physical examination, you notice significant gynecomastia, a palpable bladder, and an enlarged prostate without nodularity. Which of the following medications was he likely taking?
    a. Sildenafil
    b. Darifenacin
    c. Prazosin
    d. Dutasteride
    e. Hyoscyamine

5. A couple comes to see you regarding erectile dysfunction. The husband is 67 years old, with multiple medical comorbidities, including hypertension, stable angina, Type 2 diabetes, COPD and hypercholesterolemia. He was told by his cardiologist that he should not take any PDE5i because of one of the medications he was taking. The medication that the cardiologist was referring to most likely treats which condition this patient has?
    a. Hypertension
    b. Stable angina
    c. Type 2 diabetes
    d. COPD
    e. Hypercholesterolemia

Benign prostatic hyperplasia (BPH)

1) Pathophysiology:
    a) Age related benign enlargement of prostate
    b) Change in voiding pattern and development of voiding dysfunction (Lower Urinary Tract Symptoms (LUTS))
        i) Irritative symptoms:
            (1) Urgency, frequency, nocturia
        ii) Obstructive symptoms:
            (1) Weak stream, straining, incomplete emptying, hesitancy, intermittent stream
    c) Severe sequalae of BPH:
        i) UTI/urosepis
        ii) Urinary retention
        iii) Hydronephrosis/renal failure
        iv) Bladder calculi
        v) Gross hematuria/prostatic bleeding
    d) Very common disease in aging population
        i) Approx. 50% by age 50 and increases with increasing age (80-90% of persons with a prostate 80 years or older)
    e) Can overlap with other conditions causing LUTS:
        i) Urethral stricture
        ii) Neurologic disorders i.e. spinal cord injury, multiple sclerosis, Parkinson's disease
    f) If other causes suspected, should workup further prior to starting meds (cystoscopy, urodynamic evaluation of bladder function)
2) Mechanism of pathology twofold:
    a) Prostate volume increased from benign enlargement
        i) Mechanical outflow resistance from enlargement
        ii) Surgical therapy can de-bulk size of prostate and reduce mechanical outflow resistance and open channel
            (1) Transurethral resection of prostate (TURP)
            (2) Laser procedures (laser TURP, laser ablation/vaporization)
            (3) Minimally invasive procedures

(a) Prostatic urethral lift procedure, photoselective vaporization, water vapor thermal therapy, aquablation, prostatic artery embolization

(4) Pharmacologic agents to reduce size of prostate (5-α-reductase inhibitors (5ARI))

(a) Dihydrotestosterone (DHT) is major androgen within prostate stimulating growth

(b) DHT produced from serum testosterone within prostate by 5-α-reductase

(c) Two isoenzymes: **5-α-reductase (5AR) type I and type II**

(i) 5AR type I predominantly in lung and skin, 10% found in prostate

(ii) 5AR type II present in prostatic stroma and basal epithelial cells

(d) Principle conversion of T to DHT by 5AR type II

(e) 5AR type II most responsible for prostate growth

(f) Also indirectly affects VEGF that may contribute to prostatic bleeding from BPH

(5) 5-α-reductase inhibitors:

(a) **Finasteride (Proscar™)**

(i) Blocks 5AR type II

(ii) Reduces DHT 70-90% within prostate

(iii) Reduces PSA (prostate specific antigen by 50% at 6 months) – need to be aware and correct for prostate cancer screening

(iv) Lower dose used for androgenic baldness (**Propecia™**)

(b) **Dutasteride (Avodart™)**

(i) Blocks both isoenzymes (Type I and Type II)

(ii) Nearly eliminates DHT in prostate

(c) Both medications significantly reduce risk of urinary retention and need for surgery

(d) Also offer symptomatic improvement but takes weeks to months of treatment (unlike alpha blockers)

(e) 5ARI dosage, half-life, side effects **(see Table 1)**

(6) Combination therapy: **dutasteride** plus **tamsulosin (Jalyn™)**

b) Increased sympathetic tone of smooth muscle fibers **innervating** bladder neck and prostate

i) Autonomic innervation through adrenoreceptors/ adrenergic receptors (AR)

ii) α -1 subtype most important in urinary tract

(1) α-1a AR most concentrated in prostate (prostatic urethra and prostatic stroma) and bladder neck

(2) Agents targeting α-1 and α-1a AR can reduce sympathetic tone, promote bladder neck and prostatic urethral relaxation and alleviate symptoms (alpha blockers)

(3) Alpha blockers are competitive inhibitors of αAR

(4) Initial medications (1st generation) were non-specific alpha AR blockers (α-1 and α-2) developed for hypertension control **(phentolamine, phenoxybenzamine)**

(5) Newer medications more specific for α-1 AR (2nd generation medications **(prazosin, terazosin, doxazosin)** and α-1a AR (3rd generation alpha blockers **(alfuzosin, tamsulosin, silodosin)) (see Table 2)**

iii) Alpha blocker side effects/complications:

(1) Nonspecific alpha receptor inhibition responsible for:

(a) Cardiovascular side effects (hypotension, fatigue, dizziness, syncope) – can also have **1st dose phenomenon** with sudden transient hypotension within 30 -90 minutes

(i) 2nd generation agents **(terazosin, doxazosin)** require titration over several weeks to maximal dosage to minimize cardiovascular side effects

(ii) 3rd generation agents **(alfuzosin, tamsulosin, silodosin)** do not require dosage titration

(b) Rhinitis

(2) Retrograde ejaculation/ anejaculation

(3) Floppy iris syndrome

(a) Rare but serious complication

(b) Can make cataract surgery very difficult with increased surgical complications

(c) Has been reported with all alpha blockers

(d) Risk can remain elevated even if medication stopped

(e) May consider delaying starting alpha blocker therapy if cataract surgery planned

(f) Patients must inform ophthalmologist of alpha blocker use, and many ophthalmologists recommend ocular exam prior to initiating alpha blocker therapy

iv) Medication specific considerations

(1) **Terazosin**: absorption delayed by fatty meals with lower peak plasma concentration

(2) **Doxazosin**: hepatic metabolism and should be used cautiously in patients with liver disease

(3) **Silodosin**: dose reduction necessary in renal insufficiency (CrCL <50 mL/min)

Erectile Dysfunction (ED)

1) Pathophysiology

a) Defined as consistent inability to achieve or maintain erection for penetrative intercourse

b) May be first indicator of systemic medical disease (hypertension, diabetes, endocrinologic disorder) and young patients should be worked up for other serious medical conditions

c) Common condition and age related with increasing rates after age 40

d) Multiple possible organic, psychogenic or mixed causes:

i) Neurogenic (i.e. spinal cord injury)

ii) Hormonal (hypogonadism, treatment for prostate cancer)

iii) Vascular (hypertension, diabetes, pelvic trauma)

iv) Iatrogenic (surgery or radiation for prostate cancer or other pelvic malignancies)

v) Psychogenic (idiopathic, depression)

vi) Drug induced (typically by treatment of hypertension, prostate cancer or SSRI for depression)

vii) Lifestyle etiologies (alcohol and marijuana use/abuse)

e) Pathophysiologic mechanisms:

i) Normal erections depend on sequential cascade of psychological, neurologic, vascular and chemical processes

(1) Sexual stimulation induces parasympathetic impulse to cavernosal endothelial nitric oxide (NO) release

(2) NO induces cavernosal smooth muscle guanyl cyclase to convert cyclic GTP (cGTP) to cyclic GMP (cGMP)

(3) cGMP related phosphorylation cascade stimulates protein kinases and opens ion channels to increase cavernosal arterial flow

(4) Cavernosal engorgement causes compression of subtunical venous complex and reduces venous outflow

(5) Increased intracavernosal pressure causes rigidity for tumescence

ii) Detumescence occurs when cGMP is hydrolyzed by phosphodiesterase type 5 (PDE5) isoenzyme

iii) Eleven different PDE isoenzymes have been identified which can explain differing side effect profiles of different PDE5 inhibitors

iv) Androgens and hormonal deprivation, in addition to depressing libido, also affect cGMP and NO metabolism

f) Pharmacologic therapy for ED

i) Oral agents: PDE5 inhibitors (PDE5i) **(Table 3)**

(1) Inhibit degradation of cGMP and enhance NO effect amplifying relaxation of cavernosal smooth muscle

(2) All medications similar in efficacy but differing duration and side effects

(3) Drug class side effects include headache, facial flushing, rhinitis and dyspepsia

(4) **Sildenafil (Viagra™)** also has affinity for PDE6 which is found in the eye

(a) Can cause blue to green vision changes or blue halo

(5) **Tadalafil (Cialis™)** can cause myalgias and back pain

(6) **Vardenafil (Levitra™)** had been found to prolong the QT interval and use should be avoided with other meds increasing QT

(7) Concomitant use of nitrates or nitroglycerin contraindicated with PDE5 inhibitors because of the risk of hypotension, cardiovascular collapse and even death

(a) Elderly patients or patients with cardiovascular disease or risk factors should have a cardiology evaluation prior to starting PDE5i

(8) NAION (non-arteritic anterior ischemic optic neuropathy):

(a) Well recognized concern of all PDE5i

(b) Sudden onset of visual field defects because of irreversible optic nerve ischemia

(c) Incidence reported as high as 10/100,000

(d) Inconclusive evidence connecting NAION to PDE5i

(i) Most pts have significant cardiovascular disease or risk factors

(ii) Should ask pts about cardiovascular risk factors and visual problems before starting PDE5i

(iii) Pts should be warned to stop PDE5i immediately and go to ER if notice any change in visual acuity or fields

(9) Concomitant use with alpha blockers can also cause hypotension – recommend not to use within 2 hours of each other

ii) Local pharmacologic options for PDE5i failure **(Table 4)**

(1) Test dose for all local medications given in MD office under supervision with tutoring

(2) Injection therapies titrated up for efficacy

(3) Intraurethral medication

(a) **Alprostadil** Transurethral (**"Medicated Urethral System for Erection", MUSE™**)

(i) Synthetic prostaglandin EI (PGE1)

(ii) Stimulates adenylyl cyclase to increase cAMP levels, leading to arteriolar vasodilation, smooth muscle relaxation and increased arterial inflow for erection

(iii) Absorbed locally with minimal systemic side effects

(iv) Efficacy >60% even for patients failing PDE5i

(v) Can be combined with PDE5i

(4) Intracavernosal injection therapy

(a) **Alprastodil (Edex™, Cavarject™)**

(i) Similar to **MUSE™**

(ii) Lower dose, better efficacy (70-90%)

(b) **Papaverine** injection

(i) Nonselective PDE5i

(ii) Monotherapy: success rates of 55%

(iii) Can cause priapism or corporal fibrosis

(c) **Phentolamine** injection

(i) Alpha blocker – not effective alone for erections

(ii) Potentiates other intracavernosal injections by relaxing corporal smooth muscle and possibly increasing NO supply

(d) Combination injections

(i) Increased efficacy and lower side effects (since individual doses lowered)

(ii) Not studied or approved by FDA but available from compounding pharmacies

(iii) **Bimix**: papaverine plus phentolamine

(iv) **Trimix**: papaverine, phentolamine and PGE1

Overactive Bladder (OAB)/Lower Urinary Tract Symptoms (LUTS)

1. Bladder functions as a reservoir for storage and elimination of urine
    a. Complex neurophysiology between brain, spinal cord, autonomic pathways and GU organs
    b. During filling: lower intravesical pressure to minimize incontinence, vesico-ureteral reflux and deleterious effects of bladder dysfunction
    c. During emptying: coordinated bladder contraction, decreased urethral resistance and relaxation of the external striated sphincter
    d. Relaxation is modulated by sympathetic pathway, norepinephrine release stimulating beta receptors and bladder smooth muscle relaxation
2. Voiding (emptying) modulated by parasympathetic pathways, with detrusor smooth muscle contraction and inhibition of sympathetic tone at bladder neck and striated sphincter
3. Muscarinic (M) receptors located throughout body and GU system
    a. M2 receptor dominant form in detrusor
    b. M3 also present in bladder, as well as brain, salivary glands, colon and eye

c. During voiding, acetylcholine (Ach) interacts with M2 & M3, stimulating contraction of detrusor

4. OAB is medical condition characterized by urinary urgency, frequency, nocturia and occasionally urinary incontinence
    a. Other conditions should be ruled out: UTI, bladder cancer or neurogenic bladder dysfunction
    b. OAB theorized due to abnormal detrusor activity from Ach/M3 receptors subtypes
        i. Uninhibited bladder contractions from altered innervation from BPH, neurologic conditions (Stroke, DM, SCI) as well as idiopathic conditions
    c. Significant QOL burden and medical expenditures
5. Primary treatment of OAB are anti-muscarinic agents (**Table 5**) and beta-3 adrenergic receptor agonists**:**
    a. Competitive inhibition of M2 and M3 receptors
    b. Allow more effective storage (filling): increases bladder capacity and decreases urinary urgency
    c. Side effects related to systemic effect on muscarinic receptors:
        i. Dry mouth and constipation (most common)
        ii. Gastroesophageal reflux, cognitive impairment, blurry vision, sedation, and tachycardia less common
            1. Cognitive impairment more common in elderly/demented
                a. **Trospium (Santura™)** is a quaternary amine which is less likely to cross the blood-brain barrier and has less cognitive side effects
        iii. Contraindicated for narrow angle glaucoma and urinary retention
    d. Newer formulations (extended release, transdermal application) have less side effects due to lower peak plasma concentrations and less hepatic first pass metabolism (most are metabolized by cytochrome P450 pathway)
    e. Side effects are a common reason for noncompliance and discontinuation
6. Another mechanistic target for OAB: beta-3 adrenergic receptor agonists
    a. Found primarily in adipose (brown fat) and lipolysis
    b. Also present in bladder detrusor muscle

c. B3 adrenergic agonist improves OAB symptoms

i. **Mirabegron (Myrbetriq™)** and **Vibegron (Gemtesa™)** improve urgency, frequency and urge incontinence

ii. Less anticholinergic side effects (dry mouth, constipation, etc)

iii. Metabolized by liver (CP450) – drug interactions

Hypogonadism & Testosterone Replacement

1. Serum Testosterone (T) surges in mid-teens and then starts declining in 30's – 40's
    a. Approximately 1% per year decline, or 10% per decade after age 40
    b. Pts can be symptomatic from declining levels of T
        i. Erectile dysfunction, lower or loss of libido, depression, fatigue, decreased muscle mass, anemia, osteoporosis
    c. Diagnosis is based both on clinical findings and lab measurements
        i. Morning serum total T measurement because of normal circadian variation
    d. Hypogonadism can be primary (lack of T production by testicles) or secondary (lack of pituitary stimulation with low LH and FHS)
    e. Should have endocrine evaluation before starting T replacement
2. Testosterone replacement strategies:
    a. May benefit by increasing lean muscle mass and sexual performance/desire
    b. Can be drug of abuse for athletes
    c. Don't always have ED if hypogonadal – if patient has clinical stigmata and lab tests consistent with hypogonadism, they may benefit from T replacement
    d. Careful monitoring required, as high peak levels can suppress LH/FSH and lead to infertility
    e. Should not be used with a history of prostate cancer
3. Medications: **(Table 6)**
    a. Oral androgens traditionally not used in US due to need for daily administration, with difficulty to adequately achieve consistent therapeutic serum levels
        i. Inconsistent serum levels can cause mood swings and ED

ii. Historical oral androgens associated with hepatotoxicity

iii. Still used in other countries

iv. Approved modern oral T agents:

1. oral testosterone undecanoate (**Jatenzo™**)
    a. Twice daily with food
    b. Several dose levels (start with 158 mg, can titrate to 198 mg or 237 mg based on serum levels)
2. oral testosterone undecanoate (**Kyzatrex™**)
    a. Twice daily dosage with titration
3. oral testosterone undecanoate (**Tlando™**)
    a. Twice daily dosage with titration

b. Injectable agents:

i. Convenient dosing (every 2-4 weeks)

ii. Can have high peak and low nadir levels

c. Transdermal agents

i. Transdermal patch (**Androderm™**) and topical gels (**AndroGel™**) more closely mimic normal circadian cycle of T

ii. Can cause skin irritation and rash

iii. Other gel brands: **Fortesta™, Axiron™, Vogelxo™**

d. Buccal preparations

i. Favorable serum levels with less side effects

ii. Can cause buccal irritation and bitter taste

e. Nasal preparation

i. Intranasal gel (**Natesto™**)

ii. Quick absorption with short half-life – requires use TID

iii. Long term data lacking

iv. Can cause nasal irritation

f. Long-acting formulations

i. Long-acting injection every 6-12 weeks

ii. Implantable pellets every 6 months

Prostate Cancer Therapy with Androgen Deprivation

1. Withdrawal of androgen to castrate levels is mainstay of prostate cancer treatment
2. Androgen (primarily testosterone) is key driver of prostate health, growth and homeostasis
    a. both in combination with radiotherapy for curative therapy of localized disease
    b. palliative therapy for metastatic disease
    c. causes tumor cell apoptosis and regression
3. Decreases serum PSA levels
4. Androgen production:
    a. Interplay between hypothalamus-pituitary axis and testes
    b. Gonadotropin releasing hormone (GnRH) from hypothalamus stimulates pituitary to release luteinizing hormone to stimulate testis to produce testosterone
    c. LH binds Leydig cells in testicle for androgen production
    d. GnRH release is pulsatile (every 90 -120 min)
    e. Negative feedback of GnRH by androgen receptors in hypothalamus and pituitary
        i. Similar to home thermostat – if too hot, furnace shuts off
    f. Two classes of drugs to impact LH production **(see Table 7)**
        i. LHRH agonists – depot injection with constant elevated LH levels – negative feedback
            1. Can have initial surge in testosterone before shut off to castrate (2 weeks or greater duration of surge)
                a. Leuprolide (Lupron™, Eligard™)
                b. Goserelin (Zoladex™)
        ii. LHRH antagonist – direct competitive blockade of LH receptor in Leydig cells with immediate castration (<24 hrs)
            1. No initial surge in testosterone
                a. Degarelix (Firmagon™) injection monthly
                b. Relugolix (Orgovyx™) oral tablet daily
5. Binding of androgen to androgen receptor
    a. Several oral competitive inhibitors of androgen receptor (AR)
    b. Prevents extragonadal androgens from binding AR and stimulating prostate cancer **(see Table 8)**:
        i. Flutamide (Eulexin™)
            1. Shortest half life
            2. GI upset
        ii. Bicalutamide (Casodex™)
            1. Long half life
        iii. Nilutamide (Nilandron™)

1. Can rarely cause interstitial pneumonitis – follow CXR
2. Rare visual disturbance in low light

c. All induce cytochrome P450
   i. Can cause liver toxicity/hepatitis
      1. Follow LFTs

d. All commonly induce gynecomastia and breast tenderness/sensitivity

6. Androgen deprivation/castration analogous to menopause ("andropause")
   a. Systemic hormonal influences
      i. Hot flashes
      ii. Sexual dysfunction
      iii. Metabolic syndrome
         1. Weight gain
         2. Hypertension/cardiovascular impact
         3. Alterations in lipid panel and sugar metabolism
      iv. Osteoporosis
         1. Can mediate with Vit. D, calcium, strength training
      v. Neuro and psych disturbances
         1. Fatigue
         2. Irritability/personality change
         3. Cognitive decline
         4. Depression
7. Second generation oral hormonal therapies **(see Table 9)**:
   a. Androgen biosynthesis inhibitor
      i. **Abiraterone acetate (Zytiga™)** – approved for metastatic CRPC and initial treatment of high-volume metastatic disease
      ii. Inhibits the cytochrome P450 CYP17 enzyme important in cholesterol steroidogenesis
         1. Inhibits 17α-hydroxylase and C17-20 lyase enzymatic action of P450 CYP17, the key enzyme in androgen biosynthesis
         2. Counteracts aberrant intratumoral androgen production as well as extragonadal androgen production to maintain castrate intracellularly
         3. CYP17 also important for glucocorticoid production, with negative feedback on ACTH
            a. Co-administration of prednisone helps maintain ACTH homeostasis and avoid glucocorticoid/mineralocorticoid excess
         4. Similar mechanism as ketoconazole
      iii. Side effects/adverse reactions
         1. Hepatotoxicity and elevated LFTs

2. Mineralocorticoid excess: ↓K+, fluid retention, HTN, edema
3. Exacerbate heart disease
4. Fatigue

b. Second generation anti-androgens

i. **Enzalutamide (Xtandi™), Apalutamide (Erleada™) and Darolutamide (Nubeqa™)**

ii. Triple action and more potent than 1st generation

1. Prevents binding of ligand to AR (androgen receptors) in cytoplasm
2. Prevents translocation of activated AR to nucleus
3. Prevents binding of activated AR to DNA

iii. Indicated for non-metastatic CRPC (apalutamide and enzalutamide) and metastatic CRPC (enzalutamide) in combination with ADT

iv. **Cautions:**

1. Contraindicated with seizure disorder
2. Falls and fractures
3. Enzalutamide: ischemic heart disease
4. Apalutamide: rash (25%) and hypothyroidism (85)

v. Darolutamide: doesn't cross blood-brain barrier, less CNS concerns

1. No contraindication for seizure disorder or falls/fractures
2. Preferred for elderly, frail or unsteady gait

Table 1: Alpha blocker therapy for symptomatic BPH

Name	Side effects/notes
Second Generation Terazosin (Hytrin™)	First dose syncope; dizziness; tachycardia; headache; asthenia; rhinitis
Doxazosin (Cardura™)	Same as above
Third Generation Alfuzosin (Uroxatral™)	Dizziness; headache; minimal cardiovascular effect; ejaculatory dysfunction but less than tamsulosin
Tamsulosin (Flomax™)	Ejaculatory dysfunction; rhinitis
Silodosin (Rapaflo™)	Retrograde ejaculation

Dose titrated weekly up to desired dose

Table 2: 5-alpha reductase inhibitor medication for symptomatic BPH

Name	Half-life	Mechanism	Side effects/notes
Finasteride (Proscar™)	6-8 hours	Inhibits type II 5-AR	Decreased libido, sexual dysfunction, gynecomastia, breast tenderness
Dutasteride (Avodart™)	3-5 weeks	Inhibits type I and II 5-AR	Same as above

Table 3: Oral agents for erectile dysfunction

Name	Time to Onset	Serum Half Life	Affected by food?	Specific Side effects/notes
Sildenafil (Viagra™)	60 min	4 hours	Yes; delays onset	Visual disturbances (“blue halo”)
Vardenafil (Levitra™)	60 min	4 hours	Yes; delays onset	Increase in QT interval – avoid with other meds that increase QT
Tadalafil (Cialis™)	120 min	17.5 hours	No	Myalgia, back pain
Avanafil (Stendra™)	60 min	5 hours	No	

Table 4: Transurethral (TU) and intracavernosal (IC) therapy for erectile dysfunction

Name	Mechanism of Action	Side effects/notes
Alprostadil TU (MUSE™)	Stimulates adenylyl cyclase to ⇑cAMP	Painful erection; urethral pain or bleeding; priapism (rare)
Alprostadil IC (Cavarject™, Edex™)	Same as above	Penile pain; fibrosis; hematoma; priapism (rare)
Papaverine IC	Non-select PDE5i; ⇑cAMP & cGMP	Priapism; corporal fibrosis
Phentolamine IC	Alpha blocker; Smooth muscle dilation; ⇓ sympathetic tone	Hypotension; reflex tachycardia

Table 5: Agents for overactive bladder

Name	Side effects/notes
Antimuscarinic agents:	
Darifenacin (Enablex™)	Hepatic dosing; no renal dosing
Fesoterodine (Toviaz™)	Max 4mg daily in renal insufficiency or if also on CYP3A4 inhibitors

Hyoscyamine (Levsin™)	Less specific then newer agents with increased side effects (rarely used)
Oxybutynin immed. release (Ditropan™)	Also has antispasmodic and local anesthetic properties; Dry mouth/constipation common
Oxybutynin extend release (Ditropan XL™)	No studies in renal/hepatic impairment; slightly less dry mouth/constipation
Oxybutynin transdermal (Oxytrol™)	No studies in renal/hepatic impairment;
Oxybutynin gel, 10% (Gelnique™)	Rotate application sites to abdomen, upper arm/shoulder or legs
Solifenacin (Vesicare™)	Renal and hepatic dosing
Tolterodine (Detrol™, Detrol LA™)	Hepatic dosing
Trospium (Sanctura™, Sanctura XR™)	Renal dosing; Quaternary amine structure less likely to pass blood-brain barrier with less CNS side effects
β 3 Andrenergic Agonist:	
Mirabegron (Mybetriq™)	Hypertension – recommend BP checks after starting
Vibegron (Gemtesa™)	Urinary retention

Table 6: Testosterone Replacement

Name	Route	Notes
Testosterone cypionate (Depo-Testosterone™)	Intramuscular injection	
Testosterone enanthate (Delatestryl™)	Intramuscular injection	
Testosterone gel 1% (AndroGel™)	Topical	Apply to clean dry area on shoulder, upper arm or abdomen
Testosterone gel 1% (Testim™)	Topical	Apply to clean dry area on shoulder or upper arm
Testosterone patch (Androderm™)	Transdermal patch	Apply to clean dry area on back or arm and rotate; must remove for MRI (metallic)
Testosterone implant (Testopel™)	Implantable pellets (75 mg each)	Implant in upper buttock under local anesthesia
Testosterone undecanoate (Jatenzo™)	Oral	Titrate based on serum T levels; can cause cardiovascular events, HTN, DVT, worsening BPH or prostate cancer; can be drug of abuse (performance enhancing)
Testosterone undecanoate (Kyzatrex™)	Oral	Titrate based on serum T levels; can cause cardiovascular events, HTN, DVT, worsening BPH or prostate cancer; can be drug of abuse (performance enhancing)
Testosterone undecanoate (Tlando™)	Oral (112.5 mg pills)	Titrate based on serum T levels; can cause cardiovascular events, HTN, DVT, worsening BPH or prostate cancer; can be drug of abuse (performance enhancing)

Testosterone undecanoate (Nebido™)	Intramuscular injection	
Intranasal testosterone (Natesto™)	Nasal gel	Quick absorption; can cause nasal irritation

Table 7: Hormonal therapy for prostate cancer: LHRH agonists and antagonists

Name	Class	Administration	Note
Leuprolide (Lupron Depot™)	LHRH agonist	IM, monthly different intervals	Can cause initial testosterone surge
Leuprolide gel (Eligard™)	LHRH agonist	SC monthly, different intervals	Can cause initial testosterone surge; Requires refrigeration
Goserelin (Zoladex™and Zoladex LA™)	LHRH agonist	SC, monthly, q 3 months	Can cause initial testosterone surge
Degarelix (Firmagon™)	LHRH antagonist	Loading dose SC in 2 divided doses, Then SC monthly	No initial testosterone surge
Relugolix (Orgovyx™)	GnRH antagonist	Loading dose PO first day then PO daily	No initial testosterone surge

Table 8: Hormonal therapy for prostate cancer: Oral Anti-Androgens

Name	Half life	Administration	Note
Flutamide (Eulexin™)	6 hours	PO daily	Induces cytochrome P450 and can cause hepatitis; should follow LFTs; gynecomastia; GI upset and diarrhea
Bicalutamide (Casodex™)	7 days	PO daily	Follow LFTs; gynecomastia
Nilutamide (Nilandron™)	2 days	PO daily	Follow LFTs; gynecomastia; Rare interstitial pneumonitis – follow CXR; visual disturbances in dark

Table 9: Second generation oral agents for prostate cancer:

Name	Class	Half life	Administration	Note
Abiraterone acetate (Zytiga™)	Androgen Biosynthesis Inhibitor (ABI)	12 hours	PO daily with prednisone (take with empty stomach)	Induces cytochrome P450 and can cause hepatitis; follow LFTs;; caution with cardiac disease/CHF; can cause HTN, fluid retention and hypokalemia
Enzalutamide (Xtandi™)	2nd Gen Antiandrogen	6 days	PO daily	Induces cytochrome P450; contraindicated with seizures; association with ischemic heart disease and falls/fractures; rare PRES (posterior reversible encephalopathy; gynecomastia
Apalutamide (Erleada™)	2nd Gen Antiandrogen	3 days	PO daily	Induces cytochrome P450; Skin rash (25%); gynecomastia; contraindicated with seizures; association with falls/fractures and hypothyroidism;
Darolutamide (Nubeqa™)	2nd Gen Antiandrogen	20 hrs	PO twice daily	Induces cytochrome P450; Doesn't cross blood/brain barrier – no contraindication with seizures or falls/fractures

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. A 75-year-old patient complains of worsening urinary symptoms, including nocturia, urgency, frequency, and rare urge urinary incontinence. Urinalysis is normal, and digital rectal exam reveals an enlarged prostate, which is soft without any induration or nodularity. The patient also has a cataract, and is scheduled for cataract surgery next month. The patient wishes to try a medication to help with voiding. Which of the following medications would be contra-indicated for this patient?
    a. Tadalafil
    b. Tolterodine
    c. Tamsulosin
    d. Oxybutynin
    e. Mirabegron

    Answer: C: Tamsulosin, which is a 3rd generation alpha blocker and has been associated with complications during cataract surgery, specifically floppy iris syndrome. A is an erectile dysfunction medication which has been shown to improve BPH symptoms when taken daily in low dose; B and D are extended release antimuscarinic agent which can improve irritative BPH related symptoms; E is a beta-3 agonist, which also can improve irritative BPH related symptoms.

2. An 84-year-old nursing home resident is noted to have urinary incontinence. Urinalysis is normal, and the patient states "I can't get to the bathroom quick enough". A urodynamic evaluation reveals uninhibited bladder contractions consistent with overactive bladder. The nursing home staff put the patient in diapers, but the patient is very unhappy wearing them. The patient has had some memory loss over the past several years but is alert to person place and time. The most appropriate medication to try to alleviate the patient's symptoms is:
    a. Oxybutynin immediate release
    b. Trospium extended release
    c. Darifenacin
    d. Tolterodine
    e. Dutasteride

    Answer: B: Trospium, a quaternary amine antimuscarinic which is less likely to cross the blood-brain barrier with less cognitive side effects in patients most at risk. A, B, and D are antimuscarinics that can cross the blood-brain barrier; E is a 5-ARI which is used for BPH but not OAB.

3. A 45-year-old patient complains of gradual onset of erectile dysfunction which has worsened since starting medication for hypertension. His internist has ruled out other causes, and referred him to the urologist for evaluation and treatment. He is single, and wishes to maintain spontaneity as much as possible. Which medication will be most appropriate for him?
    a. Sildenafil
    b. Tadalafil
    c. Vardenafil
    d. Papavarine
    e. Tadalafil

    Answer: E: Tadalafil 5mg is the daily PDE5i regimen allowing the most spontaneity. A, B, and C are on-demand PDE5i, which are taken prior to every sexual encounter, and D is an intracavernosal injection which is the least spontaneous.

4. You are asked to see a 62-year-old patient in the ER for urinary retention. He is a poor historian, but does state he was taking 2 different pills for his "prostrate", the names of which he cannot remember. On physical examination, you notice significant gynecomastia, a palpable bladder, and an enlarged prostate without nodularity. Which of the following medications was he likely taking?
    a. Sildenafil
    b. Darifenacin
    c. Prazosin
    d. Dutasteride
    e. Hyoscyamine

    Answer: D: Dutasteride, a 5ARI. The most common 2 drug combination for BPH and bladder outlet obstruction is a 3rd generation alpha blocker, and a 5ARI. The 5ARI are known to cause gynecomastia. A is an ED medication; B and E are OAB medications which are used cautiously, if not at all, in BPH patients; C is a 1st generation alpha blocker which is not commonly used for BPH today because of the significant side effect profile.

5. A couple comes to see you regarding erectile dysfunction. The husband is 67 years old, with multiple medical comorbidities, including hypertension, stable angina, Type 2 diabetes mellitus, COPD and hypercholesterolemia. He was told by his cardiologist that he should not take any PDE5i because of one of the medications he was taking. The medication that the cardiologist was referring to most likely treats which condition this patient has?
    a. Hypertension
    b. Stable angina
    c. Type 2 diabetes mellitus
    d. COPD
    e. Hypercholesterolemia

    Answer: B: stable angina, for which oral nitrate medications are commonly prescribed and are contraindicated for concomitant use with PDE5i.